

Data Mining for Big Data: Map Reduce

For all questions, use a python syntax for the operations. For instance $Q = P.\text{map}(f)$ to apply a map with function f on distributed list P and store the result in distributed list Q .

1 Simple Example

node 0		
company	product	price
F	rigatoni	1
F	penne	1
C	rigatoni	5
F	tagliatelle	5
F	farfalle	9
E	rigatoni	4

node 2		
company	product	price
A	tagliatelle	9
F	canalini	5
B	bavette	3
A	canalini	7
B	canalini	8
B	tagliatelle	6

node 4		
company	product	price
D	tagliatelle	1
D	spaghetti	4
A	fusilli	6
D	canalini	7
C	fusilli	1
B	fusilli	1

node 1		
company	product	price
F	spaghetti	1
F	bavette	6
A	bavette	6
E	spaghetti	5
A	penne	6
C	spaghetti	8

node 3		
company	product	price
B	farfalle	1
C	canalini	2
A	farfalle	5
C	tagliatelle	1
D	farfalle	9
E	farfalle	8

node 5		
company	product	price
E	tagliatelle	6
E	canalini	2
E	bavette	9
B	penne	5
D	penne	6
E	penne	3

The data is represented as a list of triplets (company, product, price). The figure above shows how this list is distribution on each node.

Question 1 : We want to compute a list of pairs (company, prod_nb) representing the number of product sold by each company. Give the map/reduce operations to do it.

Question 2 : We want to compute the maximum price for each product. The output should be a list of pairs (product, max_price). Give the map/reduce operations to do it.

Question 3 : Compute the result of the map of previous question for node 2.

Question 4 : Draw a possible computation tree for product "penne".

Question 5 : During the computation of the maximum price for penne, how many pairs are send on the network without the pre-reduce optimization? and with it? (assuming the reduce for key "penne" is done on another node 6).

2 Database Operations

A database consists of two tables. The table Product(Pid, Name, Cat, Price) which contains the description of products sold. For each product:

- Pid is the identifier of the product (a number)
- Name is the name of the product (e.g., apple, orange, beer, pizza, milk, ...)
- Category is a category of product (e.g., fruit, beverage, book, clothes,...)
- Price is the price of the product.

Pid	Name	Category	Price
1	apple	fruit	1.3
2	beer	beverage	2.5
3	pizza	food	3
...			

The transactions are described in a sale table Sale(Tid, Quantity, Pid). Each transaction can contains several different products and is described by several rows with the same Tid in the Sale table.

- Tid is the transaction Id (several rows in the Sale table can have the same Tid)
- Quantity is the quantity of the product sold.
- Pid is the Id of the product sold.

Tid	Quantity	Pid
1	1.5	1
1	3	2
2	1	3
2	2	4
2	4	7
3	2	2
...		

The database is stored as lists of (key, value) pairs: a list of pairs ("Product", (Pid, Name, Cat, Price)) called P for the Product table and a list ("Sale", (Tid, Amount, Pid)) called S for the Sale table.

Question 6 : Give Map/Reduce functions to select all the products of category "book". The output must be the same format as list P .

Question 7 : Give Map/Reduce functions to output the average price of products in each category. The output should be a list of pairs (category, avg_price).

Question 8 : Give Map/Reduce functions to compute the join of the two tables on attribute "Pid". You can use a function "union" that compute the union of two distributed lists.

Question 9 : What is the replication factor for the join operation? What is the total size of the data transferred on the network? (assume Product table contains p rows and Sale table has s rows).

3 Matrix Multiplication

$$A = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 0 \end{pmatrix}; B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \end{pmatrix}$$

Assume the first solution given in the course is used for computing $C = AB$.

Question 10 : How many "Map" function calls are done on matrix A ? What are the pairs emitted by the calls on the first row of A ? How many pairs are output by these calls on the whole matrix A ?

Question 11 : During the shuffle step, give all the pairs that have a key equal to (1,3).

Question 12 : How many "Reduce" function calls are done?

Assume now the second solution is used with $g = 2$.

Question 13 : How many "Map" function calls are done on matrix A ? What are the pairs emitted by the calls on the first row of A ?

Question 14 : During the shuffle step, give all the pairs that have a key equal to (2,1).

Question 15 : How many "Reduce" function calls are done?

Assume now the third solution is used with $g = 2$.

Question 16 : In the first map step, what is block (1,2) of matrix B? By which pair is it represented in the input? What are the pairs emitted by the map call on this block?

Question 17 : During the first shuffle step, give all the pairs that have a key equal to (2,1,1). For which block of matrix C will these pairs be used?

Question 18 : How many "Reduce" function calls are done in the first step?

Question 19 : Give all the pairs that have a key of (1,1) in the second Shuffle step.

4 Word Count in Several Documents

We want to compute the number of occurrences of words in a set of document (for instance to compute tf-idf). The input consists of pairs (Did, Doc) where Did is a document id (number) and Doc is the document (a string).

Question 20 : Give Map and Reduce functions to compute pairs (word, list) where list contains (Did, count) pairs. Count is the number of time word appears in document Did. You may need to use several map/reduce steps.